



Hurricane Relief Systems for You

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Introduction

Purpose

The purpose of our hurricane relief project is to help people affected by hurricanes receive the support and resources they need quickly and safely. Our project focuses on connecting victims with volunteers, shelters, food, water, medical care, and other emergency supplies. It also helps assistance groups to organize their work with each other and communicate with people who need support. By making important information and assistance easy to find, our goal is to reduce stress and help communities recover after a hurricane.

Scope

This document will cover:

- The personas of potential users of this project
- Constraints that have been placed on this project
- A description of our project
- A competitive analysis of other systems to highlight where the project seeks to improve

Stakeholders

- Politicians
 - Local Mayors
 - Governors
- Helpers
 - First Responders
 - Police
 - Fire Fighters
 - EMT
 - Volunteers
- Residents
 - Hurricane Victims
 - Friends and Family

Marcus Rodriguez

Stats:

- Occupation: Emergency Medical Technician
- Age: 29
- Education: EMT Certification
- Tech Savviness: ★★★★★
- Technology: Iphone, Mac OS, and Emergency Radio
- LifeStyle: Works long and unpredictable hours and spends most of his free time with his family
- Hobbies: Working out, Watching football, and Fishing
- Personality: Practical, organized, dependable, and comfortable taking action in stressful situations
- From: New Orleans, Louisiana

Wish/Bother:

Wishes he had one reliable place to find everything he needs during a hurricane instead of wasting time looking through multiple sources

Quote:

“It’s seems impossible until it’s done”



Leopold Norman III

Stats:

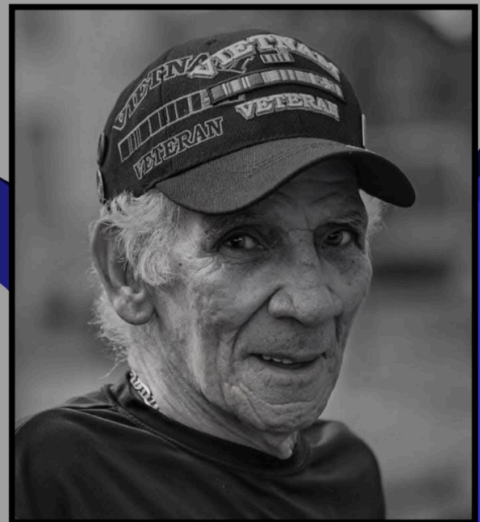
- Occupation: Cajun Restaurant Owner
- Age: 68
- Education: Highschool Diploma
- Tech Savviness: ★
- Technology: iPhone SE
- LifeStyle: Spends his time at his restaurant chatting with customers
- Hobbies: Hunting, LSU Football, and Calling Grandchildren
- Personality: Stubborn, Skeptical of new technologies, Extroverted
- From: Baton Rouge, Louisiana

Wish:

If stuck in a hurricane, he wishes to find out what he needs to do with as little interaction with his phone as possible

Quote:

“I’ve lived here all my life and have never left for a hurricane”



Jordan Mitchell

Stats:

- Occupation: Community Hurricane Relief Volunteer
- Age: 24
- Education: College Student
- Tech Savviness: ★★★★★
- Technology: iPhone and laptop
- Lifestyle: Balances school, work, and volunteering
- Hobbies: Basketball, volunteering, and spending time with friends
- Personality: Helpful, organized, hardworking, and compassionate
- From: New Orleans, Louisiana

Wish:

During a hurricane, Jordan wishes to quickly find people who need help, keep track of supplies, and communicate with other volunteers without confusion.

Quote:

“When a hurricane hits, every minute matters. I want to make sure help gets to people as fast as possible.”



Constraints

Time

- The project must be completed by December 6, 2026

Monetary

- The project must fit within the \$0 budget

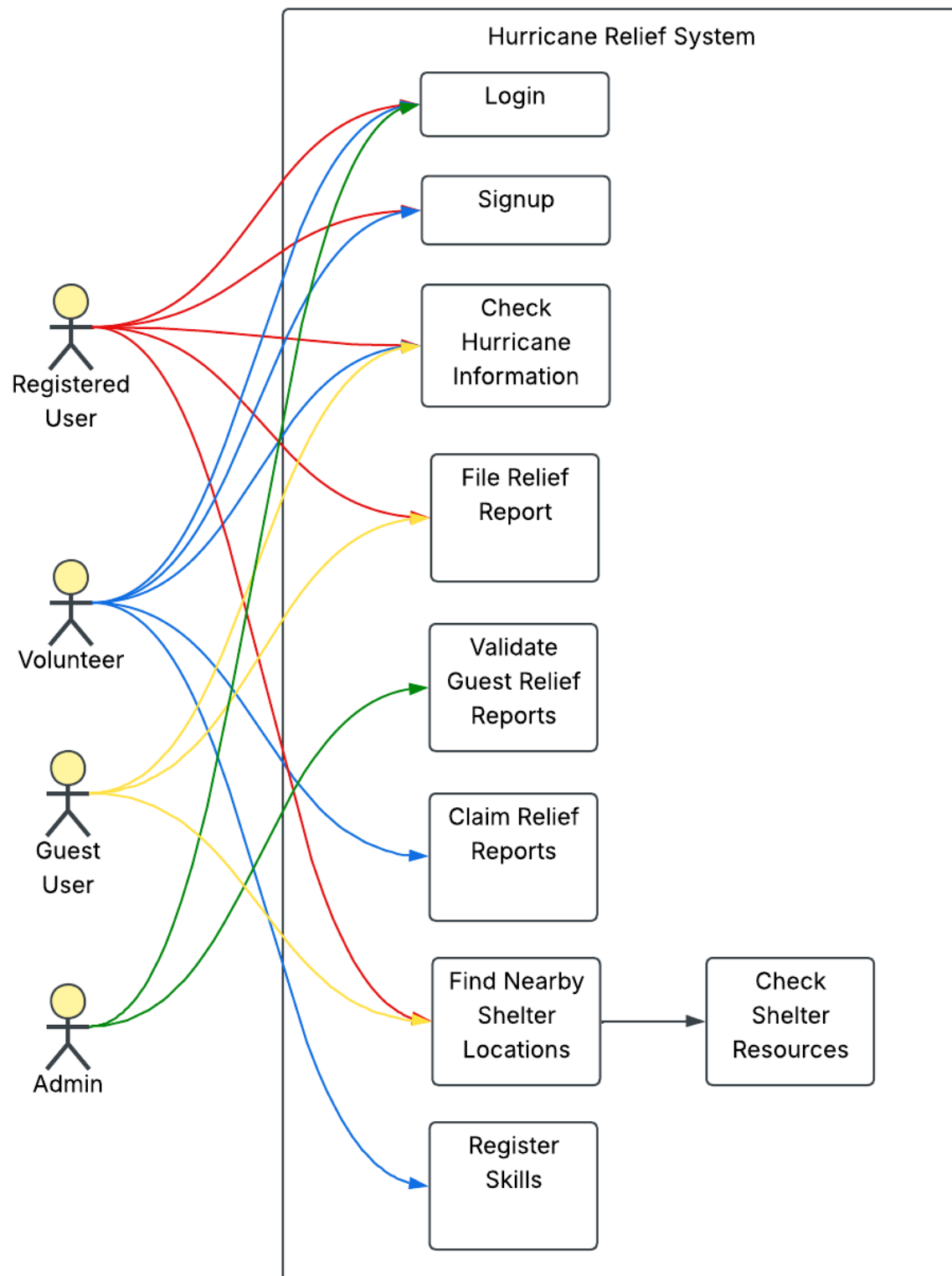
Technical

- The project must be coded in Java
- The project must include and function through a graphical user interface

Overall Description

Our hurricane relief project is a software system designed to connect hurricane survivors with the help they need after a storm. Users must be able to find nearby resource locations, medical assistance, emergency updates, and volunteer support. The system also will give volunteers a way to learn about relief opportunities and organize between different groups to find places where help is needed. By making vital information easier to access, the project will help communities to recover quickly while using a shared reporting system to connect communication between volunteers and those in need.

Business Use Cases



Functional Requirements

[Requirements Spreadsheet](#)

Requirements Spreadsheet Web Address:

https://docs.google.com/spreadsheets/d/1rKN4Nm9UstuoaeYc_fNLLkvehcsv5Y6yeMTYwICkpYE/edit?usp=sharing

Non-Functional Requirements

Look and Feel Requirements

- The user interface should be implemented in a single and easy to use way, so that distressed and unfamiliar users can easily understand how to use it.

Accessibility Requirements

- The system shall be accessible by people of all technical abilities.
- The system shall be accessible by all people with a device.

Performance Requirements

- The system shall be able to run on any device that is connected to the internet.
- The system shall be able to run with limited functionality on devices that lose internet connection.

Maintainability and Support Requirements

- The system shall be able to run on Windows, Linux, and macOS operating systems.

Security Requirements

- The user's information shall be stored in a manner that aligns with standard security procedures.

Cultural Requirements

- The system shall be accessible by both English and Spanish speaking users.

Legal Requirements

- The system shall comply with all local, state, and federal laws.

Competitive Analysis

Crisis Cleanup

Focus

- Meant to coordinate volunteer groups so that they are working efficiently and not repeating tasks.

Strengths

- It helps groups of volunteers to coordinate the separation of tasks.
- It encourages organizations to use their service as a way of coordinating volunteer efforts with other organizations
- Volunteer groups are able to claim tasks, reducing the duplication of efforts.
- It is built on a scalable architecture that can handle sudden influxes of increased traffic whenever catastrophic hurricanes occur.
- It is a web-based application, which makes it easy for lots of volunteers to access and interact with the app.

Weaknesses

- The application being web-based means that if communication services are cut-off, which is common during a hurricane, then there will be issues in groups coordinating while in the field.
- It has no authority or power, therefore, it is unable to mandate positions, assign tasks, nor distribute resources

Integrated Public Alert and Warning System

Focus

- It is a federally operated public warning system meant to give critical information to vulnerable populations on impending disasters and how to react.

Strengths

- It is a system that can push notifications through many different devices, such as radios, television, and mobile phones.

- The mobile portion of it, called Wireless Emergency Alerts (WEA), is able to give messages through a special Cell Broadcast network, allowing it to bypass possible congestion of typical communication networks. WEA is also able to give the message to any available device within a selected range without needing any information.

Weaknesses

- It only offers one way communication meaning people are unable to ask for help through it or request more information.

Veoci

Focus

- It is a software as a service platform that allows organizations to easily create and customize their own incident management systems. It is often utilized by first responders in order to organize the influx of calls and help requests immediately following a disaster.

Strengths

- Emergency services are able to quickly create custom forms without needing technical knowledge.
- It can organize geographic data so that first responders are able to focus on their jurisdiction.

Weaknesses

- It relies on cloud-based operations, meaning if a hurricane occurs and wipes out communication infrastructure, then it becomes a less valuable resource.

Summary

	Strengths	Weaknesses	Focus
<u>Crisis Cleanup:</u>	> Efficiently organizes volunteers > Coordinates between organizations	> Internet dependent	> Coordinating Volunteers

<u>Integrated Public Alert and Warning System:</u>	> Wide-reaching alerts	> Not interactive	> Alerting Potential Victims
<u>Veoci:</u>	> Ease of use for form creation > Geographic mapping	> Internet dependent	> Organizing First Responders

After competitive analysis, it is clear that there are a lot of different parties involved when it comes to hurricane response, and coordinating efficient support efforts is a complicated challenge. While these services are each able to operate well in their own domain, there is a noticeable divide between how these systems work with one another. Thus, we have decided it would be valuable for our program to provide a better form of interaction between these various groups of victims, first responders, and volunteers. With a centralized system, victims would not have to worry about whether contacting volunteer organizations or the local emergency units is the quicker route to help. Additionally, we have noted that our system must be interactable with a simple interface that allows victims to quickly and unambiguously request services and report incidents. Since hurricanes often cause damage to communication systems, it is also important that it locally saves information in the background for finding help and resources. Overall, our system aims to provide a one-stop interface for victims to receive alerts, request help, and find resources, while on the backend allowing volunteers and first responders to coordinate tasks more efficiently with one another.